

Track 3 Build Plan — Structure-Aware pHLA Stability Prediction

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TL;DR

We are building a half-life regressor that fuses a frozen protein language model with structure-derived interaction features, and we are making one specific scientific claim nobody has published: **conformational spread across a structural ensemble predicts dissociation half-life, because half-life is a kinetic quantity and every existing structural method estimates a thermodynamic one.**

The sequence-only model is our floor and it is guaranteed to work. Everything structural is upside, built on a second track that can be killed at any point without losing the submission.

What ships

Component	What it is	Status
Sequence baseline	Frozen ESM-2 embeddings + gradient-boosted head, affinity pre-trained	Must exist by 16:30 today
Structure branch	Threaded pHLA complexes, 9x20 contact tensor, ensemble mean + std	Killable at 20:00
Inverse-folding scores	ESM-IF/ProteinMPNN log-likelihood of the peptide given the pocket	Zero-shot, run early
Uncertainty	Deep ensemble + conformal prediction intervals	Non-negotiable, it is a stated gap in the field
Ablation table	seq / +contacts / +std / +IF / +termini, one Spearman each	This is the actual deliverable
Demo	Peptide + allele in, ranked half-life with error bars and the contacts that drove it	20 of 100 points

The numbers that matter

Honest state of the art on a leakage-free split is Spearman rho around **0.79**.

NetMHCstabpan reports 0.876 but was trained on its own test data, so it is not a fair target. A naive frozen-embedding baseline lands near **0.57**. Affinity pre-training alone moved ESM-2 from 0.574 to 0.745 in the published work, which makes it the highest value-per-hour thing on our list.

Anything above about 0.85 on a random split is leakage, not a result.

The biology, for ML people

Everything below exists to justify one modelling decision, so read it as feature engineering rather than as biology.

The system

An HLA class I molecule is a cell-surface protein with a groove on top. A 9-residue peptide sits in that groove. If the pair holds together long enough, the complex travels to the cell surface and a T-cell can inspect it. If it falls apart first, nothing is presented and there is no immune response. So **how long the complex survives is the thing with clinical meaning**, and it is what we predict: `thalf_hours`.

There are about 30,000 known HLA alleles. Each has a slightly different groove and therefore different peptide preferences. Our data covers 75 of them. This is why pan-allele generalisation matters and why allele-specific models are a dead end.

Anatomy you need

Each amino acid residue has two parts. The **backbone** (atoms N, CA, C, O) is identical in all 20 amino acids and forms the chain's spine. The **side chain** is everything hanging off CA, and it is the only thing that differs between amino acid types. So a protein structure is one backbone path through space, plus 20-way-variable decorations along it.

The groove has six sub-pockets, named A through F, running along its length. Pocket A holds the peptide's first residue, pocket F holds its last. The peptide's second and ninth residues typically plug into deep pockets and are called **anchor residues** - they do most of the physical work of holding the complex shut. The middle residues (4 to 7) arch upward into solvent, where a T-cell receptor can read them. Middle residues matter for recognition; anchors matter for stability.

The critical structural fact: the peptide backbone path through the groove is **near-canonical for 9-mers**. Both ends are clamped, the middle bulges, and this barely changes with sequence or allele. That is what makes our cheap structure-generation strategy

viable at all.

Why half-life is not binding affinity

This distinction is the whole project.

- **Affinity** is thermodynamic: how favourable is the bound state at equilibrium. Governed by the free energy difference, dG .
- **Stability / half-life** is kinetic: how fast does the complex fall apart. Governed by the dissociation rate, k_{off} , with half-life proportional to $1/k_{\text{off}}$.

Two complexes can have identical binding energy and very different half-lives, because the barrier to dissociation is not the depth of the well. Published work shows stability predicts T-cell immunogenicity better than affinity does.

Every existing structural method estimates the thermodynamic quantity. Rosetta energies, docking scores, co-folding confidence - all score a single static pose. None of them see dynamics. This is the gap we are aiming at, and it is why ensemble spread is our headline feature rather than a nice-to-have.

What physically sets k_{off}

A complex dissociates when thermal fluctuation carries it over a barrier. Three things resist that:

1. **The termini hydrogen-bond network.** A conserved set of HLA residues - Tyr7, Tyr59, Tyr159, Tyr171, Trp167 at the A pocket; Thr143, Lys146, Tyr84 at the F pocket - clamps the peptide's N-terminal amine and C-terminal carboxylate. Breaking these collapses stability. A hydrogen bond here is just a strong directional attraction between a donor and an acceptor atom at roughly 2.8 to 3.2 Angstroms.
2. **Anchor burial depth.** How far the P2 and P9 side chains are buried inside their pockets, measured as buried solvent-accessible surface area.
3. **Conformational entropy of the middle.** A floppy central bulge costs entropy and samples partly-unbound states; a rigid one does not.

All three are structural, all three are invisible to a sequence-only model, and all three are things we can compute in seconds once we have coordinates.

What is already done, and what is open

The sequence-plus-protein-language-model route is published. If we build only that, we are reproducing a preprint from four months ago.

The field as it stands

Work	What it does	Why it does not close our gap
NetMHCstabpan	Pan-allele neural net on half-life, the named baseline to beat	Sequence only, black box, reported scores inflated by training on its own test data
TLStab	Transfer learning from affinity and eluted ligand data	Heavy peptide overlap with its test set
SPEARMINT preprint, June 2026	Protein language model to half-life with leakage-aware splits, assay conditioning	Exactly our brief, already answered. Sequence only. No uncertainty. Treats the zero floor as a real value
3pHLA-score, 2022	Rosetta energies decomposed per peptide position, random forest per allele	Predicts affinity, not stability. Needs PyRosetta. Only 28 alleles, not pan-allele
pHLA-RF, 2020	Residue-residue contact features from 155k modelled structures, pan-allele random forest	Binary classifier using eluted ligands as a stability proxy, not half-life regression. 300,000 CPU-hours to build

Our thesis in one paragraph

Structure-based pHLA work has always predicted the wrong quantity. pHLA-RF and 3pHLA-score both produce a score for a single static pose, which approximates binding free energy. Half-life is kinetic. We take the cheap, pan-allele contact featurisation from pHLA-RF, restore the per-position resolution from 3pHLA-score without needing Rosetta, generate a small ensemble of poses per complex, and use the **variance** of the features across that ensemble as a proxy for local conformational entropy. Variance is the kinetic signal. A sequence model cannot see it and a single-structure model cannot see it either.

Four open gaps we can credibly address in 24 hours

1. **No uncertainty quantification anywhere in the stability literature.** The challenge brief and the primer both ask for it. Conformal prediction intervals are a few hours of work and nobody has them.
2. **The zero floor is treated as a real measurement.** 5,679 of our 28,166 rows are exactly 0.0 - that is a detection limit, not a half-life of zero. The published work feeds $\log(1+0) = 0$ straight into MSE. A two-head classify-then-regress formulation is defensible and novel here.

3. **No structure anywhere.** The primer lists it as an approach; nobody in the stability literature has done it.
4. **Interpretability.** Our features are indexed by physical contact, so every prediction decomposes into contacts we can render. NetMHCstabpan cannot do this and neither can a language model head.

We do not need all four. Two done properly beats four done badly.

The data, and four traps in it

28,166 rows, 5 columns, no nulls, no duplicate (allele, peptide) pairs. 75 alleles, 5,633 distinct peptides, **every peptide exactly 9 residues**. Target `t_half_hours`: median 1.1, mean 5.45, max 256.7, raw skew 5.87.

This is the SPEARMINT paper's stage-2 corpus (they report 27,034 rows / 72 alleles). Their setup is directly reproducible and their Table 2 is effectively our scoreboard.

Trap 1: a random row split leaks, badly

Each peptide appears with a median of 4 alleles, up to 36. 70% of peptides appear with more than one allele, and **94% of rows involve such a peptide**. A random split puts the same peptide on both sides and inflates every metric.

What we do: cluster unique peptides at 80% sequence identity using normalised Levenshtein distance, then assign **whole clusters** to train/val/test. Target proportions roughly 21,600 / 2,700 / 2,700, matching the published split so our numbers are comparable to theirs.

This is the single highest-risk correctness item in the project. If the split is wrong, every number we report is wrong, and a judge who asks "how did you split" will find it in thirty seconds.

Trap 2: 20% of labels are exactly zero

5,679 rows at exactly 0.0. That is an assay detection floor, not a measurement of zero half-life. Silently applying $\log_{1p}$ treats a censoring boundary as a real value, which is what the published work does.

What we do: **two heads**. A classifier for $P(t_half = 0)$ and a regressor trained only on the non-zero rows. Final prediction is $P(\text{nonzero}) * E[t_half \mid \text{nonzero}]$. Simpler than a censored likelihood, defensible in Q&A, and it gives us a second metric axis for free.

Trap 3: the target needs a transform

$\log_{10}$ drops skew from 5.87 to 0.91. Train in log space with MSE, report in hours by inverting with $\exp(y) - 1$. The published work uses exactly this, so we stay comparable.

Trap 4: the pseudo-sequence cannot separate two alleles

`hla_pseudoseq` has only 74 distinct values for 75 alleles. `HLA-B*14:01(C67S)` and `HLA-B*14:02(C67S)` share one, producing 368 colliding (peptide, pseudoseq) pairs.

Use `hla_seq`, the 182-residue sequence, or the allele name as identity. Never key on the pseudo-sequence. Note that this collision is exactly the failure mode our alignment-free contact featurisation avoids by construction, so it is worth a sentence in the pitch.

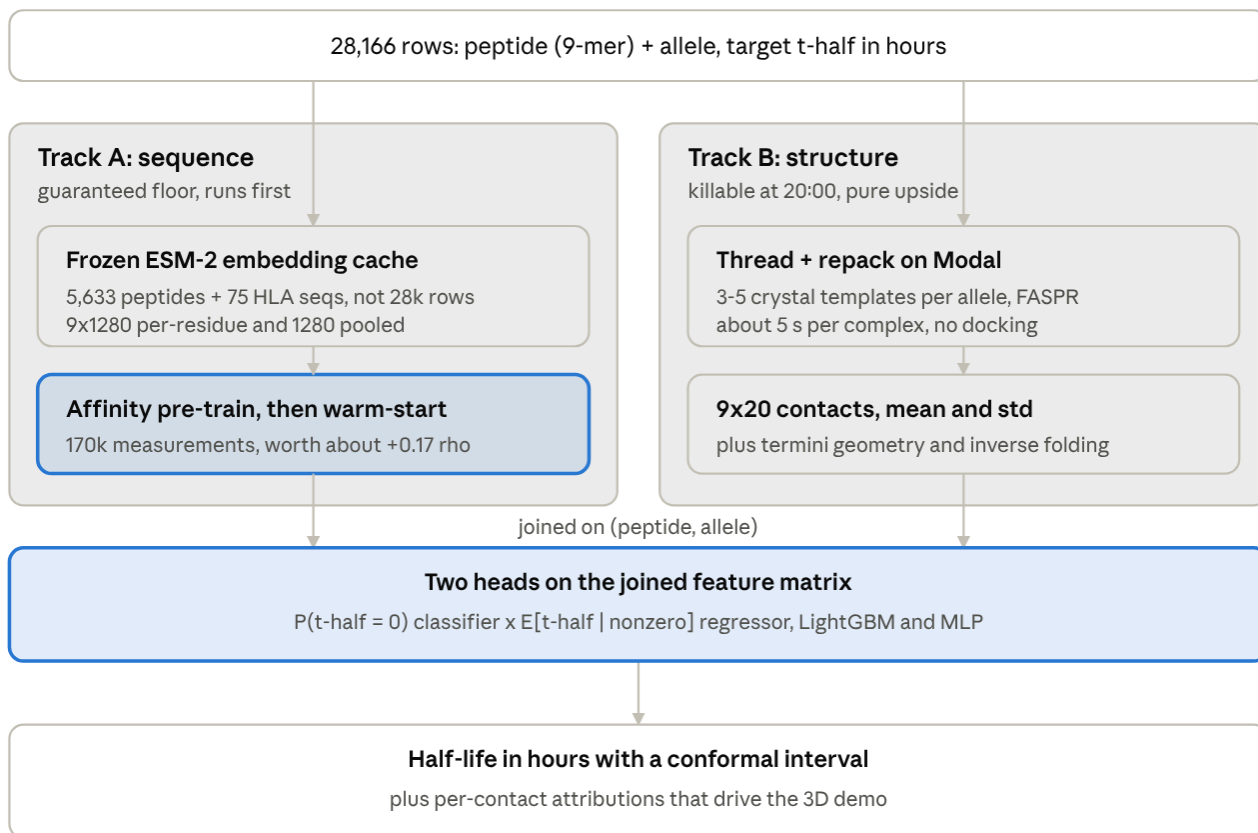
External data we also want

- **NetMHCstabpan training data** from the DTU service page (training data table plus `data.tar.gz`). Download now, it is the cleanest source of half-life labels and lets us check our split against theirs.
- **The affinity corpus** for pre-training: NetMHCpan 4.1, about 170,107 measurements over 109 alleles, labels as $1 - \log(\text{IC}_{50}) / \log(50000)$.
- **PDB crystal structures** of HLA class I complexes, for threading templates.

System architecture

Everything expensive is computed once, then the head trains in seconds

Two tracks that never block each other, joined on (peptide, allele)



system architecture · two tracks, one join, two heads

The design principle is that **everything expensive is computed once**. The embedding cache and the threaded structures are both precomputed and written to disk; after that the head trains in under a minute, which is the difference between running three experiments and running fifty.

The two tracks share only the join key. Track A can ship without Track B at any point.

Track A: the sequence backbone

Do not fine-tune the backbone. With 24 hours, fine-tuning 650M parameters is how we end up with nothing. Freeze it, precompute embeddings once, train a small head. Everything expensive runs once; then the head trains in under a minute and we iterate fifty times instead of three.

Step A1: the caching trick

We have 28,166 rows but only **75 distinct HLA sequences** and **5,633 distinct peptides**. Embed the unique entities, not the rows.

```
for each of 5,633 peptides: ESM-2 forward -> (9, 1280) per-residue + (1280,)
mean-pooled
for each of 75 HLA seqs:   ESM-2 forward -> (1280,) mean-pooled
row lookup:                dict keyed by sequence string
```

5,708 forward passes instead of 28,166. Minutes on one GPU, not hours. Save to `.npy` and never recompute.

We lose cross-chain attention by embedding the two chains separately. Accept that. The published gain from cross-chain attention was +0.158 but it requires joint forward passes over all 28k rows, and we can add joint embedding later if time allows.

Step A2: the first number, by 16:30

Concatenate `[peptide mean-pool (1280) || peptide per-residue flattened (9x1280) || HLA mean-pool (1280)]`, train LightGBM on log1p target, evaluate Spearman on the clustered test split. This is the floor. It should land somewhere near 0.57 based on published ESM-2 direct numbers.

If the per-residue flattening is too wide for LightGBM, reduce with PCA to 256 dims per position first. Do not spend more than twenty minutes on this.

Step A3: affinity pre-training - the highest value-per-hour item

Published result: ESM-2 went from 0.574 to 0.745 by adding affinity pre-training. That is **+0.171 from a curriculum change, more than the entire architecture upgrade gave.**

On frozen embeddings this is cheap:

1. Embed the affinity corpus using the same per-entity cache (many peptides and alleles overlap with the stability set, so the marginal cost is small).
2. Train the head on affinity with target `1 - log(IC50)/log(50000)`.
3. Warm-start the same head on stability. Keep the pre-trained head weights rather than re-initialising - the published work found that keeping them beats re-initialising.

For LightGBM, "warm start" means initialising with the affinity model and continuing training on stability with a lower learning rate. For the MLP it is literal weight transfer.

Decontaminate across tasks. The affinity set and the stability test split must not share peptide clusters, or we have reintroduced leakage through the back door. Run the same 80% identity clustering across the union of both corpora.

Step A4: what to try only if there is slack

- ESM-C instead of ESM-2, if an API key is in the sponsor credits
- Joint peptide+HLA sequence in one forward pass with a chain-ID tensor, no separator token - this is what MINT does and it is where the cross-chain gain comes from
- Attention-pooling over residues instead of mean-pooling

None of these is on the critical path.

Track B: generating structures without docking

We need one structure per (peptide, allele) pair - about 28,000 of them, times 3 to 5 templates each. Docking is impossible at that scale. Threading is the answer and it takes seconds.

Why we do not dock

APE-Gen, the pHLA-specific modelling tool used by both reference papers, takes about **2 minutes per complex on 6 cores**. The pHLA-RF dataset of 155,562 structures cost roughly **300,000 CPU-hours**. For our 28k pairs that is 933 CPU-hours, which is survivable on Modal fan-out but puts the whole Saturday night on one install working.

Docking solves "where does the ligand go and in what conformation". For 9-mers in a class I groove, that is already solved: the backbone path is near-canonical. We are paying for a search whose answer we already know.

Threading, precisely

Threading means taking a known-correct backbone from a crystal structure and swapping in our sequence's side chains, keeping the backbone coordinates exactly as they are.

Template 3MRE (HLA-A*02:01 bound to GILGFVFTL):

position	1	2	3	4	5	6	7	8	9
template	G	I	L	G	F	V	F	T	L
our peptide	N	L	V	P	M	V	A	T	V

keep N/CA/C/O coordinates at every position
delete old side chains, graft new ones on

The name is literal: the sequence is threaded through a fixed structural scaffold, like beads onto a wire.

What it costs: the backbone is assumed, not computed. A peptide with a genuinely unusual backbone will not be found. For 9-mers this is rare enough to accept. For 10-mers and longer it degrades - not our problem, our dataset is 100% 9-mers.

Repacking, precisely

Grafting a side chain raises an immediate question: which way does it point? Side chains rotate about their single bonds; the angles are called **chi angles** (chi-1 about the CA-CB bond, chi-2 about CB-CG, and so on). Leucine has 2, lysine has 4, alanine has none.

Surveying thousands of crystal structures shows these angles cluster into a handful of preferred low-strain conformations. Each cluster is a **rotamer**, and **rotamer libraries** tabulate them. This turns a continuous optimisation into a discrete one: pick one rotamer per position from a short list.

Repacking is choosing the rotamer at every position so side chains pack without clashing. The choices are coupled - if P2's leucine swings left it blocks the space P3's valine wanted - so it is a combinatorial search, not a per-position decision.

We use **FASPR**: fast, open source, no licence key, about 0.1 seconds for a complex this size. PyRosetta's packer is better and far slower and needs a registered licence. Skip it.

Repack the groove too, not just the peptide. HLA pockets are not rigid - a bulky P2 tryptophan pushes groove residues aside. Repacking only the peptide produces artificial clashes and junk contact features. Include HLA residues within about 6 Angstroms of the peptide.

The full per-pair job

```
input: peptide NLVPMVATV, allele HLA-A*02:01

1. look up 3-5 crystal templates for that allele
2. for each template:
    thread - keep HLA atoms; keep peptide backbone;
           delete peptide side chains; set sequence
    repack - FASPR picks rotamers for the 9 peptide side chains
           plus groove residues within 6 A
3. output: 3-5 complete all-atom complexes, ~5 s each
```

At 5 to 10 seconds per complex and 3 templates, 28k pairs is roughly 230 CPU-hours - under an hour wall-clock with 256-way parallelism on Modal.

Templates: the gating check, do this first

Before anything else in Track B, count how many of our 75 alleles have a direct crystal structure in the PDB versus needing a homology model. This single number decides whether threading is a one-hour or a four-hour job.

- Query RCSB for HLA class I entries, parse the allele from the entity metadata, map to our 75.
- Alleles with direct entries: use 3 to 5 of their own structures as templates.
- Alleles without: use a template from the same **supertype** - a grouping of alleles with similar groove chemistry and binding preferences. Slightly worse, still usable.
- HLA-A*02:01 alone has dozens of PDB entries, so the well-covered alleles give us ensembles for free.

Modal

The structure fan-out and the embedding pass both run on Modal. That earns Challenge 1 ("Best use of Modal") honestly rather than as a bolt-on, and it is the only way the 28k generation finishes overnight. One container image with FASPR and Biotite baked in, a `.map()` over pairs, results written to a volume as compressed coordinate arrays rather than PDB text.

Featurisation: turning coordinates into a vector

Four feature blocks, each independently ablatable. Build them in this order.

B1: the 9x20 contact tensor

This is our combination of the two reference papers, and the exact form is not published anywhere.

What pHLA-RF did: 210 numbers, one per unordered amino-acid-type pair ($C(20,2) + 20 = 210$). For slot {LEU, PHE}, find every peptide leucine against every HLA phenylalanine (and vice versa), measure the **nearest heavy-atom distance** for each pair, apply a decreasing function, sum.

Heavy atom means non-hydrogen - hydrogens are not resolved in crystal structures. Nearest-atom, not centre-to-centre, because side chains reach: a lysine amine can be 6 Angstroms from its own CA.

Why that is clever: it is indexed by interaction chemistry, not by residue position, so it needs no alignment and no pseudo-sequence. Any allele, any mutant, maps into the same space. Allele identity is implicit in which contacts physically form. This is exactly the

property NetMHCstabpan fakes with a hand-picked 34-residue pseudo-sequence, and it is why our Trap 4 collision cannot happen to us.

Why it is not enough: it throws positional information away completely. A LEU-PHE contact at P2, the anchor holding the complex shut, counts identically to one at P5 pointing into solvent. Survivable for binary classification, fatal for half-life.

What we build instead: index by peptide position and HLA residue type.

For peptide position p in 1..9 and HLA amino-acid type Y in the 20 types: sum $f(d)$ over all contacts between the residue at p and every HLA residue of type Y .

9 x 20 = 180 features. Positional on the peptide side, alignment-free on the HLA side (we only ever name the *type* of an HLA residue, never its position). No Rosetta, no licence, pure coordinates.

The soft contact function

There is no threshold. Every residue pair contributes a continuous amount between 0 and 1:

$$f(d) = \frac{1}{1 + e^{\lambda \cdot 2(d - 5)}}$$

distance	contact value	reading
3.5 Å	0.95	van der Waals contact
4.5 Å	0.73	in contact
5.0 Å	0.50	the midpoint
6.0 Å	0.12	fading
8.0 Å	0.002	effectively nothing

A hard 5 Angstrom cutoff would make 4.99 count fully and 5.01 count zero. Our structures are models with roughly 1 Angstrom coordinate error, so a knife-edge turns modelling noise into feature noise. The sigmoid saturates at both ends and only the genuinely ambiguous 3.5-to-6.5 band is sensitive. The published ablation found the sigmoid beat $1/d$ and $1/d$ -squared for exactly this reason: $1/d$ diverges at short range so one near-clash swamps the sum, and its tail is long so distant residues still contribute.

Optional upgrade if time allows: three distance shells (under 4, 4-6, 6-8 Angstroms) giving $9 \times 20 \times 3 = 540$ features. This is the hand-rolled version of radial basis expansion in a 3D GNN.

```

import numpy as np
from scipy.spatial import cKDTree
AAs = "ACDEFGHIKLMNPQRSTVWY"

def featurise(pep, hla):          # each: list of (restype_char,
heavy_coords[n,3])
    F = np.zeros((9, 20))
    trees = [(AAs.index(t), cKDTree(c)) for t, c in hla]
    for p, (_, pcoords) in enumerate(pep):
        for y, tree in trees:
            d = tree.query(pcoords)[0].min()    # nearest heavy-atom distance
            F[p, y] += 1.0 / (1.0 + np.exp(2.0 * (d - 5.0)))
    return F.ravel()                # 180 features

```

Do not restrict to anchor positions. The published ablation showed anchors-only roughly equals all-nine, so the signal is there - but restricting costs us and buys nothing. Keeping all nine separate is free, a tree learns to ignore the noisy middle by itself, we keep allele-specific exceptions like HLA-B*08:01 where position 5 genuinely is an anchor, and the learned feature importances become a figure in the pitch: "we never told it about anchors, here it is rediscovering them."

B2: ensemble mean and std - the headline feature

For each of the 180 features, compute **mean and standard deviation across the 3-5 threaded poses**. That gives 360 features.

The mean block is the ordinary structural signal. **The std block is the claim.**

Half-life is $1/k_{\text{off}}$. A peptide whose contacts barely change across several real crystallographic backbones is rigidly locked and dissociates slowly. One whose contacts scatter is loosely held, samples partly-unbound states, and dissociates fast. Standard deviation across the ensemble is a cheap proxy for local conformational entropy, and it is something a sequence-only model structurally cannot see.

Using **real crystal templates** rather than synthetic perturbation matters here: the spread reflects genuine observed backbone variation, not noise we invented.

Indirect support: pHLA-RF's AUROC went 0.975 to 0.990 when trained on the full conformational ensemble rather than the single best pose.

The caveat a sharp judge will raise: that paper also notes APE-Gen produced about 5 more conformers for true binders than non-binders, so conformer *count* leaks the label. Defend by fixing the number of poses per complex at a constant, or including count as an explicit feature and reporting its importance. Pre-empting this in Q&A is worth more than the extra accuracy.

B3: termini and anchor geometry - features built for stability specifically

About 20 features that affinity predictors do not prioritise, targeting the three physical drivers of k_{off} :

- Hydrogen-bond count and geometry from the peptide's N-terminal amine and C-terminal carboxylate to the conserved clamp residues (Tyr7, Tyr59, Tyr159, Tyr171, Trp167, Thr143, Lys146, Tyr84). A hydrogen bond here is a donor-acceptor pair at 2.8 to 3.2 Angstroms with reasonable angle.
- Buried solvent-accessible surface area of the P2 and P9 side chains. SASA is the area a water-sized probe can reach; buried SASA is the residue's free SASA minus its SASA in the complex. Compute with Shrake-Rupley, available in Biotite.
- SASA of the central bulge (P4-P7), as an entropy-cost proxy.
- Bulge height: maximum displacement of the central CA atoms from the line joining the two anchor CAs.

This is the block that lets us say in the pitch: we built features for stability, we did not borrow features from affinity prediction.

B4: inverse folding - a foundation model reading geometry

ESM-IF1 or ProteinMPNN takes a backbone and outputs, per position, a distribution over which amino acid belongs there. Score our actual peptide against it:

```
s_p = \log P(\text{residue}_p \mid \text{pHLA backbone})
```

This is literally "how well does this residue fit this pocket", answered by a protein foundation model conditioned on 3D structure, with zero training and one forward pass. It is the cleanest possible answer to the track's own question. Decoding is autoregressive so we get per-position logits for free - that is a 9-vector that slots straight into the per-position framing.

Run this tonight on 500 pairs before committing to anything heavier. If raw inverse-folding log-likelihood correlates with half-life at all, that alone is a slide, and it is a zero-shot result we can report before any head is trained.

Variance across decoding orders or temperatures gives another ensemble - in sequence-likelihood space rather than coordinate space, and nearly free.

The prediction head

Two heads, because of the zero floor

```
P(t-half = 0)          <- binary classifier, all rows
E[t-half | t-half>0] <- regressor, non-zero rows only, log1p target

prediction = (1 - P_zero) * (exp(y_hat) - 1)
```

Train them on the same feature matrix, separately. Report both the regression metrics on non-zero rows and the classifier AUROC, plus the combined metric on everything. This handles the censoring honestly without needing a Tobit likelihood, and it gives us a second axis to show improvement on.

Two model classes, trained in parallel

LightGBM is the primary. It handles our wide heterogeneous feature matrix, it is fast enough to retrain in under a minute so we can ablate aggressively, and it gives per-prediction feature attributions for free - which is what the demo runs on.

A 2-layer MLP is the secondary, with a Gaussian likelihood head predicting mean and log-variance. This gives aleatoric uncertainty directly and is what we ensemble for epistemic uncertainty.

Train both, report both. If they disagree substantially, that disagreement is itself informative about which feature blocks are carrying signal.

Uncertainty - do not skip this

The SPEARMINT paper has no uncertainty quantification anywhere. The challenge brief and the primer both ask for it. This is cheap and nobody else will have it.

Three layers, in order of cost:

1. **Deep ensemble.** Train 5 MLPs with different seeds. Mean is the prediction, variance across members is epistemic uncertainty. About 10 minutes total on frozen features.
2. **Conformal prediction intervals.** Take the absolute residuals on the validation split, find the 90th percentile, and emit `prediction +/- q`. This gives a **distribution-free guarantee**: on exchangeable data, the true value falls inside the interval 90% of the time. Twenty lines of code.
3. **Normalised conformal**, if there is time: scale the interval by the ensemble's predicted standard deviation so intervals are wide where the model is unsure and tight where it is confident. Much better looking in the demo than a constant-width band.

Report the empirical coverage on the test split. If we claim 90% intervals and observe 89%, that is a calibration result and it is a slide. If we observe 60%, we say so - a miscalibration we measured is better than one we did not look for.

Metrics to report

Metric	Why
Spearman rho	Primary. Assays disagree on absolute values but rank consistently
Pearson r on log	Catches the NetMHCstabpan failure mode: it ranks at 0.876 but its Pearson is only 0.532, so its absolute hours are badly calibrated
RMSE in log space	Absolute accuracy
Precision@10	The clinical use is re-ranking a shortlist, so top-k is the metric that matches the application
Interval coverage at 90%	Nobody else reports this
Per-allele Spearman	Shows robustness, and exposes whether we are just fitting the data-rich alleles
Held-out-allele Spearman	The real generalisation test, and where structure should win

That last row is where our story lives. Beating NetMHCstabpan outright on the main split is hard. Beating it **on alleles it has never seen** is both achievable and more scientifically interesting, because that is exactly what a pan-allele structural method should be good at.

Evaluation: the ablation table is the deliverable

A leaderboard number proves nothing on its own. A clean ablation on a leakage-free split proves we know which part of our system does the work, and that is what earns the technicality points.

The table we fill in

Every row is the same head on the same split, differing only in which feature blocks are concatenated. Fill it top to bottom as blocks come online.

Features	Spearman	Pearson	Held-out allele	Notes
Peptide sequence only (one-hot or BLOSUM)				the dumbest possible floor
ESM-2 frozen, no transfer				expect roughly 0.57
+ affinity pre-training				expect the biggest single jump
+ inverse-folding log-likelihoods				zero-shot structure signal
+ contact tensor, mean only				ordinary structural signal
+ contact tensor, mean and std				the claim: does spread add signal
+ termini and anchor geometry				stability-specific features
NetMHCstabpan on our split				the honest comparison

The row that matters most is mean-only versus mean-and-std. That is a falsifiable test of our one original hypothesis, and it is a single number either way.

Split discipline

- Cluster peptides at 80% identity by normalised Levenshtein, assign **whole clusters** to train/val/test.
- Decontaminate the affinity pre-training corpus against the stability test clusters.
- Fix the split seed and commit the split indices to the repo. Every ablation row uses the identical split.
- Hold out a set of **whole alleles** as a second test axis, never seen in training.

Reporting the negative result

If the std block adds nothing, **we report that**. A clean negative with a proper ablation on a leakage-free split is a legitimate finding, it is honest, and it plays far better in Q&A than a vague claim nobody can check. The ablation table makes that outcome respectable rather than embarrassing.

Comparability

We match the published setup deliberately - same $\log(1+t)$ transform, same 80% identity clustering, same roughly 21.6k/2.7k/2.7k proportions - so our numbers can be placed next to their Table 2 without caveats. If we diverge from their setup anywhere, say so explicitly in the writeup.

The demo

Demo is 20 of 100 points, a full fifth of the score. A regression table is not a demo. Budget real time for this and assign an owner by 18:00.

What is on screen

1. **Input:** paste a peptide, pick an allele from a dropdown. Or paste a protein sequence and we tile all its 9-mers.
2. **Output:** a ranked shortlist with predicted half-life in hours and a calibrated interval on each row. Sorted, because the clinical use is re-ranking a candidate list.
3. **The thing nobody else has:** click a row and the 3D complex renders with the contacts that drove that prediction highlighted - green for stabilising contributions, red for destabilising, thickness by magnitude.
4. **A caption generated from the attributions:** "Predicted 4.2 h (90% interval 2.1 to 7.8). 31% of the positive contribution came from P2 leucine burial against the B-pocket phenylalanines."

Why this wins points the model alone does not

NetMHCstabpan is a black box and so is a language-model head. Interpretability is the thing structure buys us that scale does not. "Here is *why* it said 4.2 hours" is a different conversation from "here is our scatter plot", and it is the natural answer to the sponsor's framing about protein design workflows - a designer needs to know which residue to change.

Technical notes

- Attributions come free from LightGBM via `treeinterpreter` or SHAP. Because our features are indexed by (peptide position, HLA residue type), every contribution maps back to a physical contact we can draw.
- 3D rendering: NGL Viewer or 3Dmol.js in the browser, fed the threaded PDB. No server-side rendering needed.
- Precompute everything for a handful of demo peptides so the live demo cannot fail on a cold GPU. Have a fallback recording.

- Pick demo peptides with a story: a known cancer neoantigen, a viral epitope, and a pair that differ at one anchor position so you can show the prediction flip. Mutating P2 from a good anchor to tryptophan should visibly collapse the predicted half-life - that is the single most convincing 15 seconds available to us.

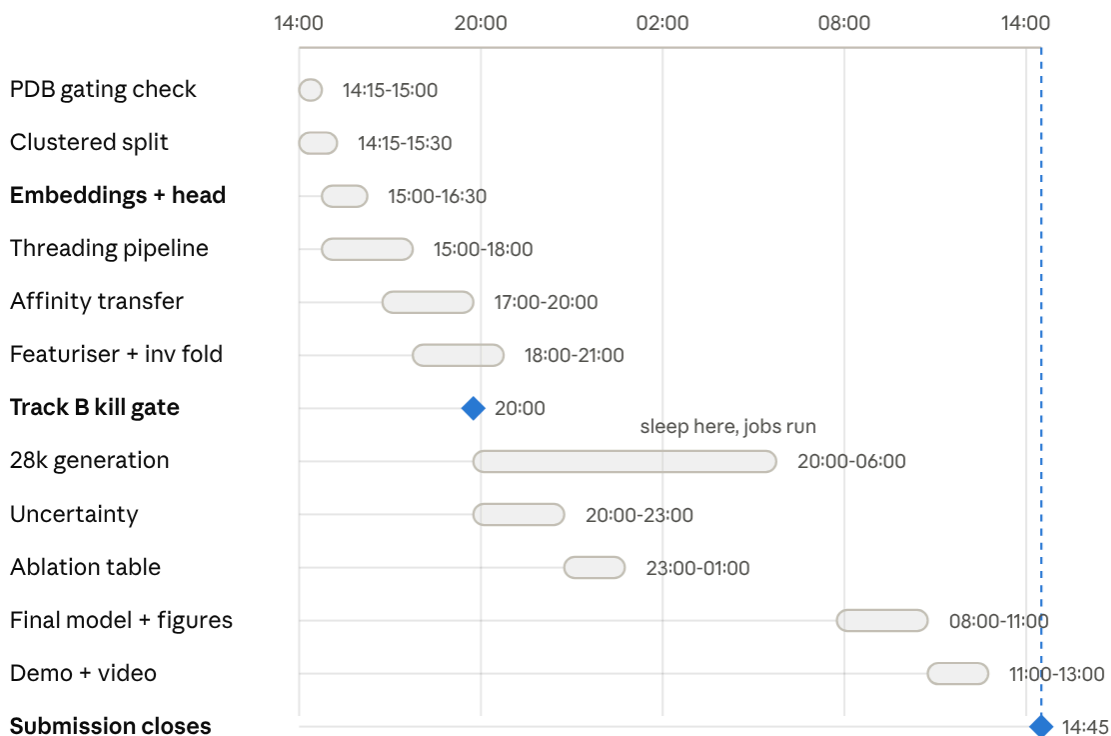
Pitch structure, 1:30

- 15s: stability predicts immunogenicity better than affinity, and it is what decides whether a neoantigen is worth making.
- 20s: everyone predicts it from sequence. Structure has been tried, but always for the wrong quantity - affinity is thermodynamic, half-life is kinetic.
- 25s: our move. Cheap threading gives us an ensemble of real poses per complex; the *variance* of interaction features across that ensemble is a proxy for how rigidly the peptide is held, which is what sets the dissociation rate.
- 20s: the ablation table, one line on the number.
- 10s: hand to demo.

Timeline

A Spearman number by 16:30, or the plan is already behind

Saturday 14:15 to Sunday 14:45, structure track killable at 20:00



build timeline · 14:15 Saturday to 14:45 Sunday, one kill gate

Two hard checkpoints. **16:30 today: a Spearman number from the sequence model.** If that has not happened, everything structural stops until it does. **20:00 today: the Track B kill gate.** If threading is not producing valid structures by then, we kill it and ship what Track A has.

The overnight window is the only reason the 28k generation is feasible at all, so the Modal job must be launched before anyone sleeps. Everything after 08:00 Sunday is assembly, not research - no new ideas after that line.

Risks and kill criteria

The governing rule: **the sequence-only model is our floor and it is guaranteed.**
Everything structural is upside. Build in that order and we cannot end up with nothing.

Risk	Kill criterion	Fallback
Threading pipeline not producing valid structures	Not working by 20:00	Kill Track B entirely. Ship sequence model plus inverse folding on whatever structures we have. Still a structure story
28k x 3 generation too slow overnight	Not half done by 02:00	Stratified 6k-pair subset. Show the structure branch adds signal there, ship the sequence model on full data. Scoped result, stated plainly
Structure features add nothing	Ablation shows no gain	Report it. A measured negative on a leakage-free split is a legitimate finding and good Q&A material
Many alleles have no PDB template	Gating check comes back poor	Supertype homology templates, or restrict the structure branch to covered alleles and report on that subset
FASPR or Biotite install fights us	More than 45 minutes lost	Drop repacking entirely; use backbone plus CB positions only. Contact features degrade but still work

Risk	Kill criterion	Fallback
Split is subtly wrong	Any metric above about 0.85 on the main split	Stop and audit immediately. That number is leakage, not a result
Demo breaks live	-	Precomputed predictions for demo peptides, plus a recorded fallback video
Someone fine-tunes a 650M model "quickly"	At any point	Veto. We agreed not to

The two things that will actually sink us

Time sunk into installs. PyRosetta, APE-Gen, MODELLER - each is a plausible four-hour hole. The plan deliberately avoids all three. If anyone starts installing PyRosetta, stop them.

A wrong split discovered at 13:00 Sunday. Build the split first, commit the indices, and have a second person sanity-check that no peptide cluster appears on both sides. Thirty minutes now, saves the submission.

What we say if things go badly

An honest, well-scoped partial result with a correct evaluation beats an impressive-sounding number nobody can reproduce. If we end up with "sequence model at 0.74, structure branch tested on 6k pairs showed no significant gain, here is the ablation and here is why we think the ensemble was too small" - that is a real scientific contribution to a 24-hour hackathon and it will survive Q&A. A 0.9 from a leaky split will not.

Division of labour and interfaces

The two tracks must not block each other. They meet at one file format, agreed now.

The interface contract

Everything is keyed on (peptide, allele). Every workstream writes a parquet file with those two columns plus its features, and the merge is a join. Nobody waits for anybody.

data/splits/clusters.parquet	peptide, cluster_id, split
data/emb/peptide_esm2.npy	5633 x (9x1280 + 1280), + index.json
data/emb/hla_esm2.npy	75 x 1280, + index.json
data/struct/{allele}/{peptide}.npz	coords, per template
data/feat/contacts.parquet	peptide, allele, f_mean_000..179, f_std_000..179
data/feat/termini.parquet	peptide, allele, ~20 cols
data/feat/invfold.parquet	peptide, allele, s_1..s_9, s_std_1..s_9

Agree these column names before anyone writes code. A rename at 02:00 costs an hour.

Roles

Workstream	Owns	First deliverable
Data and splits	Clustered split, zero handling, the parquet contract, decontamination	<code>clusters.parquet</code> by 15:30
Sequence model	Embedding cache, LightGBM and MLP heads, affinity transfer, uncertainty	A Spearman number by 16:30
Structure generation	PDB template survey, threading, FASPR, Modal fan-out	One valid complex by 16:00, 200-pair batch by 18:00
Featurisation	The 9x20 featuriser, termini geometry, inverse folding	Featuriser tested on the 200-pair batch by 19:00
Demo and writeup	Front end, 3D viewer, attribution rendering, video, repo hygiene	Wireframe by 20:00, working by 11:00 Sunday

With fewer people, merge the last two into whoever finishes first. The data-and-splits role cannot be shared - one person owns the split and everyone else consumes it.

Repo hygiene, because it is submitted

- One repo, clean README with the ablation table in it, requirements pinned.
- **No prior commits.** Rule 1 of the submission rules: built entirely during the event.
- Credit every pre-trained model and dataset: ESM-2, ESM-IF/ProteinMPNN, FASPR, the NetMHCstabpan data, the PDB entries we template from. Open-source models are allowed if properly credited.
- Commit the split indices and the exact ablation script so a judge can rerun one row.

Working agreements

- Nobody fine-tunes a backbone. Agreed in advance.
- Nobody installs PyRosetta, APE-Gen or MODELLER without the group agreeing to the time cost.
- Any metric above 0.85 on the main split gets audited before it gets celebrated.
- Drafts of the pitch and demo exist by 20:00 tonight, not 11:00 tomorrow.

Glossary

Allele - one specific variant of the HLA gene, e.g. HLA-A*02:01. About 30,000 are known; our data covers 75.

Anchor residue - a peptide residue whose side chain plugs deep into a groove pocket, usually position 2 and position 9 in a 9-mer. Anchors do most of the physical work of holding the complex together.

Angstrom (Å) - 0.1 nanometres. A carbon-carbon bond is about 1.5 Å; residues are "in contact" at roughly 4 to 5 Å.

Backbone - the N, CA, C, O atoms common to every amino acid, forming the protein chain's spine.

Buried SASA - a residue's solvent-accessible surface area when free, minus its SASA in the complex. How much of it got hidden by binding. A direct measure of burial depth.

Chi angle - a rotation angle about a side-chain single bond. Chi-1 is about the CA-CB bond, chi-2 about CB-CG, and so on.

Conformational ensemble - the set of distinct shapes a molecule actually samples at thermal equilibrium, as opposed to the single lowest-energy structure a predictor hands you.

Co-folding - predicting the structure of a complex directly from the sequences of both chains (AlphaFold3, Boltz-2, Chai-1). Expensive; we are not using it on the critical path.

Desolvation - the energetic cost of burying a polar group away from water. One of the main terms in any protein force field.

Eluted ligand (EL) - a peptide recovered from the cell surface by mass spectrometry. Used as a proxy for "this complex was stable enough to reach the surface".

FASPR - a fast open-source side-chain packer. Backbone plus sequence in, complete structure out, about 0.1 s.

Groove / pockets A-F - the peptide-binding cleft on top of an HLA class I molecule, divided into six sub-pockets along its length. Pocket A holds the peptide's first residue, pocket F its last.

Heavy atom - any non-hydrogen atom. Hydrogens are not resolved in crystal structures, so all contact definitions use heavy atoms.

HLA (human leukocyte antigen) - the human MHC. Class I molecules present peptides to CD8 T-cells.

Hydrogen bond - a strong directional attraction between a donor (N-H, O-H) and an acceptor (O, N) at roughly 2.8 to 3.2 Å.

Inverse folding - given a backbone, predict which amino acid sequence would fold into it. ESM-IF1 and ProteinMPNN do this; scoring a known sequence against the output gives "how well does this residue fit this position".

k_{off} - the dissociation rate constant. Half-life is proportional to $1/k_{\text{off}}$. This is the quantity we are predicting, and it is kinetic, not thermodynamic.

Pan-allele - a single model that works for any allele, including ones absent from training. Opposed to per-allele models.

pHLA / pMHC - a peptide-HLA complex. Our prediction target is a property of this object.

PDB - the Protein Data Bank, the public archive of experimentally determined structures. Fewer than 1,000 pHLA crystal structures exist, which is why modelling is necessary at all.

Pseudo-sequence - a hand-picked 34-residue subset of the HLA sequence covering groove contact positions, used by NetMHC-family models to represent the allele compactly. Our data has 74 distinct pseudo-sequences for 75 alleles, hence the collision.

Repacking - choosing a rotamer for every side chain so they pack together without clashing. Combinatorial, because the choices are coupled.

Rosetta / ref2015 - a widely used protein force field. `ref2015` is its current all-atom energy function, with about 19 terms. PyRosetta is the Python binding; it needs a free academic licence and we are avoiding it.

Rotamer - a commonly observed low-strain side-chain conformation. A **rotamer library** tabulates them, turning a continuous optimisation into a discrete choice.

SASA (solvent-accessible surface area) - the area of a molecule reachable by a water-sized probe. Computed with the Shrake-Rupley algorithm.

Scoring function - a weighted sum of physical and statistical energy terms, used to estimate how favourable a structure is.

Side chain - everything hanging off CA, starting at CB. The only part that differs between amino acid types.

Supertype - a grouping of HLA alleles with similar groove chemistry and overlapping peptide preferences. Lets us borrow a structural template across alleles.

Threading - placing a sequence onto an existing backbone from a template structure, keeping the backbone coordinates fixed.

t-half (thalf_hours) - the dissociation half-life of the complex in hours. Our regression target.